Hybrid experiment theoretical essentials

Alexei Gurchenko

April 2026

Contents

1 Standing wave optical cavity in vacuum **2**
1.1 Reflection and transmission at an interface . . . 2
1.2 Reflected, transmitted and intracavity fields and intensities . . . 3
1.2.1 Reflected field and intensity . . . 3
1.2.2 Transmitted field and intensity . . . 4
1.2.3 Intracavity field and intensity . . . 4
1.3 Measuring cavity parameters . . . 5
1.4 Mode matching . . . 6

2 Cold atoms and atom-light interaction **7**
2.1 Energy levels . . . 7
2.2 Atoms cooling and trapping . . . 7
2.2.1 Magneto-optical trap . . . 7
2.2.2 Optical molasses . . . 7
2.3 Atom-light interaction . . . 9
2.3.1 Effective Hamiltonian . . . 9
2.3.2 One-axis twisting . . . 9

3 Optomechanics **9**
3.1 Cavity optomechanics . . . 9
3.1.1 Linearized Hamiltonian . . . 9
3.1.2 Optical spring . . . 9
3.1.3 Sideband cooling . . . 9
3.2 Zigzag cavity . . . 9

4 Hybrid system description and dynamics **9**

1 Standing wave optical cavity in vacuum

1.1 Reflection and transmission at an interface

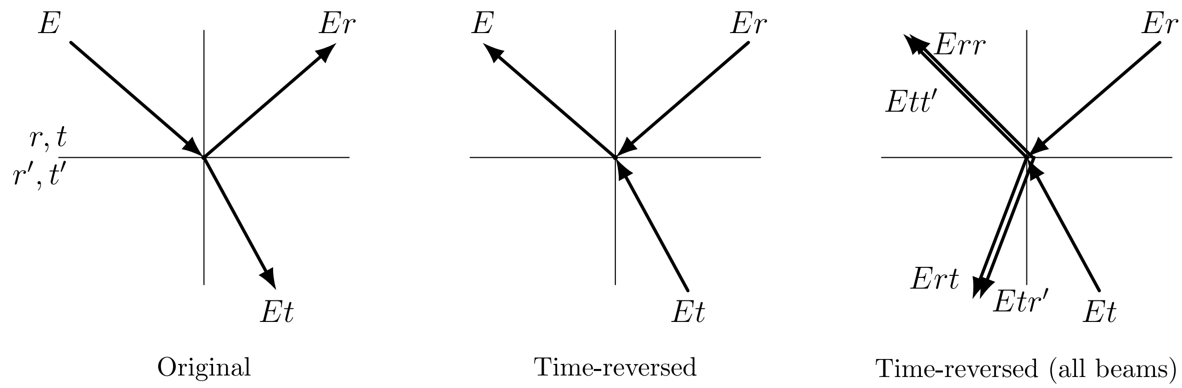

Figure 1: Original and time-reversed interaction between a beam and an interface

The first question we need to answer before moving on to calculating reflected, transmitted and intracavity fields, is the following. If light is shined to an interface from one side and the interface has reflected coefficient r and transmitted coefficient t, what would be the reflected and transmitted coefficients r' and t' if the light is shined from another side? This elegant derivation [1] of the relation between the field transmission and reflection coefficients was made by G. G. Stokes. He assumed that there was no dissipation at the interface and therefore the process obeyed time-reversal invariance. The three waves are shown on the left in the figure 1. The time-reversed case appears in the center plot. When we reverse time, we need to include three additional waves: the reflection from the bottom wave, and the transmission of both the bottom and top waves. All of the waves are shown in the plot on the right. In order for the right-hand plot to be equivalent to the time-reversed center plot, the two waves in the upper left must sum to E and the two waves in the lower left must sum to zero. Thus,

$$E = Ett' + Err \tag{1.1}$$
$$0 = Ert + Er't \tag{1.2}$$

from which we conclude that

$$t' = \frac{1 - r^2}{t} \tag{1.3}$$
$$r' = -r \tag{1.4}$$

Here we assume that $r, r', t, t' \in \mathbb{C}$. The fact, that we assumed no dissipation should mean, that

$$|t|^2 + |r|^2 = 1 \tag{1.5}$$
$$|t'|^2 + |r'|^2 = 1 \tag{1.6}$$

Let's plug 1.4 and 1.3 to 1.6. After some algebra we arrive to

$$\frac{1}{|t|^2}(|1 - r^2|^2 - (1 - |r|^2)^2) = 0 \tag{1.7}$$

Since $|r| \leq 1$, $(r = |r|e^{i\varphi_r})$ it should mean that

$$|1 - r^2| = 1 - |r|^2 = \sqrt{1 - 2|r|^2 \cos(2\varphi_r) + |r|^4} \tag{1.8}$$

Equation 1.8 is satisfied $\forall\ |r| \leq 1$ only if $\varphi_r = 0$ or π. That means that $r, r' \in \mathbb{R}$. Then $(t = |t|e^{i\varphi_t})$

$$t' = \frac{|t|^2}{t} = |t|e^{-i\varphi_t} = t^* \tag{1.9}$$

So finally we have

$$r' = -r, \quad r, r' \in \mathbb{R} \tag{1.10}$$
$$t' = t^*, \quad t, t' \in \mathbb{C} \tag{1.11}$$

1.2 Reflected, transmitted and intracavity fields and intensities

Let us start calculating the reflected, transmitted and intracavity fields. Figure 2 depicts the schematic of the electromagnetic wave traveling inside the optical cavity. We assume that the mirrors 1 and 2 have corresponding amplitude transmission and reflection coefficients r_1, t_1 and r_2, t_2. The intracavity medium is lossless, however there is a loss of amplitude due to scattering at the mirror surfaces: t_a at Mirror 1 and t_b at Mirror 2 ($t = t_a t_b$). This model is the closest approximation of a two-mirror cavity inside ultra-high vacuum environment.

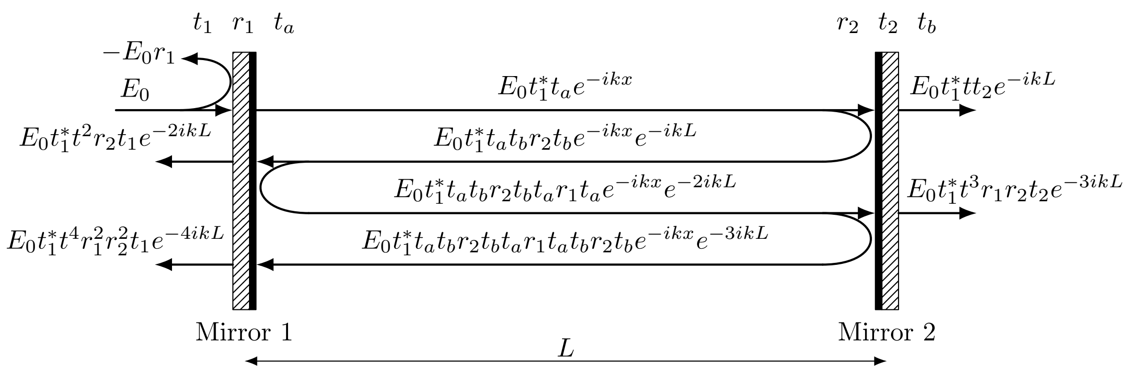

Figure 2: Optical cavity schematic

1.2.1 Reflected field and intensity

If the incident amplitude of the field is E_0, by summing all the paths that contribute to the reflected field and using 1.10, 1.11 we arrive to the following series (we relabel $t = t_a t_b$)

$$\begin{aligned} E_r &= E_0(-r_1 + t_1^* t r_2 t t_1 e^{-2ikL} + t_1^* t r_2 t r_1 t r_2 t t_1 e^{-4ikL} + ...) = \\ &= E_0(-r_1 + |t_1|^2 t^2 r_2 e^{-2ikL}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...)) = \\ &= E_0(-r_1 + \frac{|t_1|^2 t^2 r_2 e^{-2ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}}) = E_0 \frac{t^2 r_2 e^{-2ikL} - r_1}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

We obtain an expression for the reflected field:

$$E_r = E_0 \frac{t^2 r_2 e^{-2ikL} - r_1}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.12}$$

Let's remember that

$$FSR = \frac{c}{2L} \tag{1.13}$$
$$kL = \frac{2\pi L f}{c} = \frac{\pi f}{FSR} = \pi \tilde{f} \tag{1.14}$$

where $\tilde{f} = \frac{f}{FSR}$ is a normalized frequency. After some algebra we get an expression for reflected intensity:

$$I_r = I_0 \frac{(r_1 - t^2 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.15}$$

The minimum reflected intensity is obtained on resonance (when $\tilde{f} \in \mathbb{N}$):

$$I_r^{res} = I_0 \frac{(r_1 - t^2 r_2)^2}{(1 - t^2 r_1 r_2)^2} \tag{1.16}$$

1.2.2 Transmitted field and intensity

If the incident amplitude of the field is E_0, by summing all the paths that contribute to the transmitted field and using 1.10, 1.11 we arrive to the following series (we relabel $t = t_a t_b$)

$$\begin{aligned} E_t &= E_0(t_1^* t t_2 e^{-ikL} + t_1^* t r_2 t r_1 t t_2 e^{-3ikL} + ...) = \\ &= E_0 t_1^* t t_2 e^{-ikL}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...) = \\ &= E_0 \frac{t_1^* t t_2 e^{-ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

We obtain an expression for the transmitted field:

$$E_t = E_0 \frac{t_1^* t t_2 e^{-ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.17}$$

After some algebra we get an expression for transmitted intensity ($\tilde{f} = f/FSR$):

$$I_t = I_0 \frac{|t_1|^2 |t_2|^2 t^2}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.18}$$

The maximum transmitted intensity is obtained on resonance (when $\tilde{f} \in \mathbb{N}$):

$$I_t^{res} = I_0 \frac{|t_1|^2 |t_2|^2 t^2}{(1 - t^2 r_1 r_2)^2} \tag{1.19}$$

1.2.3 Intracavity field and intensity

We need to calculate fields propagating in the direction of incident light ($E_c^+(x)$), in the opposite direction ($E_c^-(x)$) and full intracavity field ($E_c(x)$). The process of calculation is very similar to the reflected and transmitted fields calculations, but this time we have to take care of at which point in the cavity the field is being calculated. Let's pick a point at a distance x from the first mirror and calculate forward and backward fields there. For the forward field:

$$\begin{aligned} E_c^+(x) &= E_0(t_1^* t_a e^{-ikx} + t_1^* t_a t_b r_2 t_b t_a r_1 t_a e^{-ikx} e^{-2ikL} + ...) = \\ &= E_0 t_1^* t_a e^{-ikx}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...) = \\ &= E_0 \frac{t_1^* t_a e^{-ikx}}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

We obtain:

$$E_c^+(x) = E_0 \frac{t_1^* t_a e^{-ikx}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.20}$$

The forward propagating intensity is:

$$I_c^+ = I_0 \frac{|t_1|^2 t_a^2}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.21}$$

For the backward field:

$$\begin{aligned} E_c^-(x) &= E_0(t_1^* t_a t_b r_2 t_b e^{-ik(2L-x)} + t_1^* t_a t_b r_2 t_b t_a r_1 t_a t_b r_2 t_b e^{-ik(4L-x)} + ...) = \\ &= E_0 t_1^* t r_2 t_b e^{-ik(2L-x)}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...) = \\ &= E_0 \frac{t_1^* t r_2 t_b e^{-ik(2L-x)}}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

It's here that we have to acknowledge that the amplitude reflection coefficients are actually negative, because upon reflection from a medium with higher refractive index the light undergoes a π phase shift. So if we take this into account we obtain:

$$E_c^-(x) = E_0 \frac{-t_1^* t |r_2| t_b e^{-ik(2L-x)}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.22}$$

Full intracavity field:

$$E_c(x) = E_c^+(x) + E_c^-(x) = E_0 \frac{t_1^* t_a e^{-ikx} - t_1^* t |r_2| t_b e^{-ik(2L-x)}}{1 - t^2 r_1 r_2 e^{-2ikL}}$$

We obtain:

$$E_c(x) = E_0 \frac{t_1^* (t_a e^{ik(L-x)} - t t_b |r_2| e^{-ik(L-x)}) e^{-ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.23}$$

After some algebra we get an expression for intracavity intensity ($\tilde{f} = f/FSR$, $\tilde{x} = 2x/\lambda$):

$$I_c(\tilde{x}) = I_0 \frac{|t_1|^2((t_a + t t_b |r_2|)^2 - 4t^2 |r_2| \cos^2(\pi(\tilde{x} - \tilde{f})))}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.24}$$

The maximum internal intensity is obtained on resonance (when $\tilde{x} = m + 1/2, \quad \tilde{f}, m \in \mathbb{N}$):

$$I_c^{res,max} = I_0 \frac{|t_1|^2 (t_a + t t_b |r_2|)^2}{(1 - t^2 r_1 r_2)^2} \approx I_0 \frac{4|t_1|^2}{(1 - t^2 r_1 r_2)^2} \tag{1.25}$$

The propagating internal intensity on resonance:

$$I_c^{res,+} = I_0 \frac{|t_1|^2 t_a^2}{(1 - t^2 r_1 r_2)^2} \approx I_0 \frac{|t_1|^2}{(1 - t^2 r_1 r_2)^2} \tag{1.26}$$

1.3 Measuring cavity parameters

Parameter of the cavity, characterizing the "goodness" of the cavity is called *finesse* ($\mathcal{F}$). It's defined as follows:

$$\mathcal{F} := \frac{FSR}{\gamma} \tag{1.27}$$

where γ is full-width at half-maximum (FWHM) of transmitted or internal intensities. If we now equate

$$I_t(\tilde{f}_{1/2}) = \frac{I_t^{res}}{2} \tag{1.28}$$

After some algebra and series expansions (since $1 - t^2 r_1 r_2 \ll 1$) we get:

$$\tilde{f}_{1/2} = \frac{1 - t^2 r_1 r_2}{2\pi} = \frac{\gamma}{2FSR} \tag{1.29}$$

$$\mathcal{F} = \frac{\pi}{1 - t^2 r_1 r_2} \tag{1.30}$$

Finesse is also related to the intensity enhancement inside of the cavity:

$$\frac{I_c^{res}}{I_0} = 4 \frac{1 - r_1^2}{(1 - t^2 r_1 r_2)^2} \tag{1.31}$$

In a lossless and symmetric ($r_1 = r_2, \ t = 0$) case it simply yields:

$$\frac{I_c^{res}}{I_0} = \frac{4\mathcal{F}}{\pi} \tag{1.32}$$

Please take into account that "lossless and symmetric ($r_1 = r_2,\ t = 0$) case" is never actually achievable in real life, and the results 1.32 for ideal case and 1.31 for a real case vary **significantly**, sometimes **by several orders of magnitude**.

If one represents $|t|, |r_1|, |r_2|$ as follows:

$$|r_1| = 1 - a,\ a \ll 1 \tag{1.33}$$

$$|r_2| = 1 - b,\ b \ll 1 \tag{1.34}$$

$$|t| = 1 - c,\ c \ll 1 \tag{1.35}$$

it becomes possible to make series expansions of all the quantities derived above. After some algebra it turns out that if one experimentally measures finesse, reflected intensity on resonance and transmitted intensity on resonance, intracavity intensity, mirror parameters and cavity losses can be derived from these three parameters with simple expressions:

$$\frac{I_c^{res}}{I_0} = \frac{4\mathcal{F}}{\pi}(1 - \sqrt{R_c}) \tag{1.36}$$

$$R_1 = 1 - \pi\frac{1 - \sqrt{R_c}}{\mathcal{F}} \tag{1.37}$$

$$R_2 = 1 - \pi\frac{T_c}{\mathcal{F}(1 - \sqrt{R_c})} \tag{1.38}$$

$$A_{loss} = 1 - T_a T_b = \pi\frac{1 - R_c - T_c}{2\mathcal{F}(1 - \sqrt{R_c})} \tag{1.39}$$

A_{loss} here is a single-pass loss coefficient. If we want to calculate the overall steady-state loss coefficient its given by a trivial expression (which is consistent with the equations above):

$$A_{loss}^{full} = 1 - R_c - T_c \tag{1.40}$$

1.4 Mode matching

Usually, depending on how well-aligned the beam is to the cavity mode, the focusing parameters of the beam, the quality of collimating lens, there is a coupling efficiency that represents the proportion of the beam intensity, coupling into the desired mode of the cavity (usually TE_{00}).

$$\eta = \frac{I_{00}}{I_0},\ I_0 = I_{00} + I_h \tag{1.41}$$

The rest of the intensity (I_h) is partly decomposed into high-order Hermite-Gaussian modes of the cavity, and partly plainly reflected from the mirror, because they cannot be decomposed into the high-order modes. The statement, that an arbitrary beam can be decomposed into HG modes is only true for the beams of the same pointing (propagating along the same axis), so if we have a beam thats misaligned with the cavity axis, there will inevitably be a part of the beam that cannot be decomposed into HG modes.

Taking into account the coupling efficiency η we can express apparent reflectance and transmittance of the cavity through the actual ones ($R_c = I_r^{cav}/I_{00},\ R_c^{app} = I_r^{full}/I_0,\ T_c = I_t^{cav}/I_{00},\ T_c^{app} = I_t^{full}/I_0$):

$$R_c^{app} = \eta R_c + 1 - \eta \tag{1.42}$$

$$T_c^{app} = \eta T_c \tag{1.43}$$

And reverse:

$$R_c = \frac{1}{\eta}(R_c^{app} - 1 + \eta) \tag{1.44}$$

$$T_c = \frac{1}{\eta}T_c^{app} \tag{1.45}$$

2 Cold atoms and atom-light interaction

2.1 Energy levels

The experiment that will be described here will work with ^{87}Rb atoms. Specifically with the D2 line of ^{87}Rb (Fig. 3). We are mostly interested in addressing the hyperfine ground states $|g_\uparrow\rangle = |5^2S_{1/2}, F = 2, m_F = 0\rangle = |\uparrow\rangle$ and $|g_\downarrow\rangle = |5^2S_{1/2}, F = 1, m_F = 0\rangle = |\downarrow\rangle$ (so-called Rb clock states). The selection rules for electric dipole (E1) transitions say that if $\Delta L = 0$ the transition is forbidden [3]. The $|\uparrow\rangle$ and $|\downarrow\rangle$ states have the same angular momentum number $L = 1$, hence the E1 transition between them is forbidden. It is not forbidden for the magnetic dipole (M1) transitions to happen between these two levels. That's why it is possible to drive this transition with a microwave antenna. Since there is a manifold of excited states $5^2P_{3/2}$ separated by an optical frequency ($\lambda = 780$ nm) from the ground $5^2S_{1/2}$ states, the system can be driven into these excited states via a laser. From now on the hyperfine levels of $5^2S_{1/2}$ will be referred to as $F = 1, 2$ and hyperfine levels of $5^2P_{3/2}$ will be referred to as $F' = 0, 1, 2, 3$. Each of the F and F' levels have a Zeeman manifold which splits the level into $2F - 1$ states. Let's for now only consider magnetically insensitive states with $m_F = 0$. There is a selection rule that says that for E1 transitions with $\Delta F = 0$, $m_F = 0 \not\leftrightarrow 0$, which means that transitions $|F = 1, m_F = 0\rangle \not\leftrightarrow |F' = 1, m_F = 0\rangle$ and $|F = 2, m_F = 0\rangle \not\leftrightarrow |F' = 2, m_F = 0\rangle$ are forbidden.

2.2 Atoms cooling and trapping

2.2.1 Magneto-optical trap

2.2.2 Optical molasses

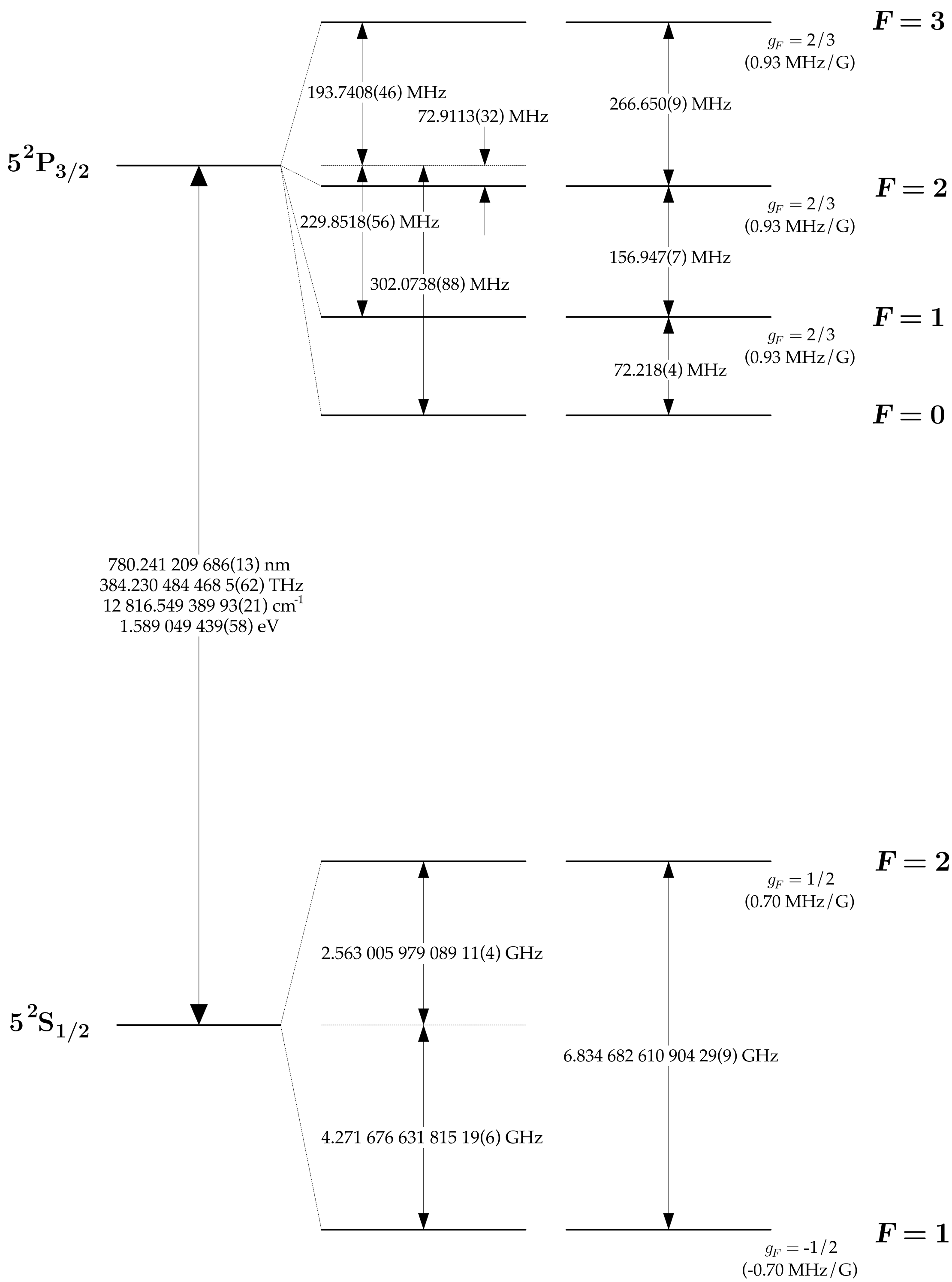

Figure 3: ^{87}Rb level diagram [2]

2.3 Atom-light interaction

The most physically accurate way do describe atom-light interaction for a ^{87}Rb ensemble driven with π-polarized light is to treat a Rubidium atom as two not interacting two-level systems 4. In case of D2 line of ^{87}Rb $|g_A\rangle = |5^2S_{1/2}, F = 2, m_F = 0\rangle = |\uparrow\rangle$, $|g_B\rangle = |5^2S_{1/2}, F = 1, m_F = 0\rangle = |\downarrow\rangle$, $|e_A\rangle = |5^2S_{3/2}, F = \{1,3\}, m_F = 0\rangle$, $|e_B\rangle = |5^2S_{3/2}, F = \{0,2\}, m_F = 0\rangle$ [2].

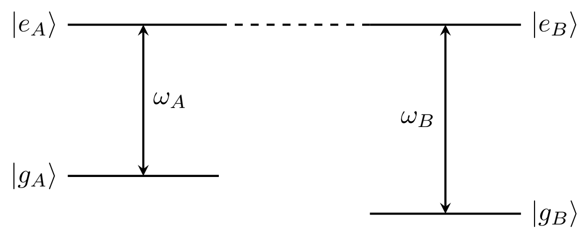

Figure 4: Level diagram

2.3.1 Effective Hamiltonian

2.3.2 One-axis twisting

3 Optomechanics

3.1 Cavity optomechanics

3.1.1 Linearized Hamiltonian

3.1.2 Optical spring

3.1.3 Sideband cooling

3.2 Zigzag cavity

4 Hybrid system description and dynamics

References

[1] Warren Nagourney. *Quantum Electronics for Atomic Physics and Telecommunication*. 2014. DOI: 10.1093/acprof:oso/9780199665488.001.0001.

[2] Daniel Steck. "Rubidium 87 d line data, revision 1.6". In: *Source–http://steck. us/alkalidata* (2003).

[3] C.J. Foot. "Atomic Physics (Oxford Master Series in Atomic, Optical and Laser Physics)". In: *Citeulike.Org* 11 (2005).